

Weekly Progress Report – Lightweight Linux GUI Distros for Small-Scale Private Cloud Deployment

This report summarises the progress completed during Week 3 and Week 4 of *Examination of Lightweight Functional Linux GUI Distros for Small-Scale Private Cloud Deployment project*. The purpose of this report is to demonstrate weekly continuity, practical implementation, and baseline testing progress.

Week 3 – Private Cloud Deployment Installation Notes

Docker was successfully installed on Lubuntu, Xubuntu, and Linux Lite without major compatibility issues. The installation process was largely identical across all three distributions due to their Ubuntu-based architecture. No significant dependency conflicts or installation errors were encountered during the setup process.

Following the Docker installation, a nginx container was deployed on each virtual machine to simulate a lightweight private cloud service. Functionality was verified by accessing the default nginx web page through a web browser, confirming that the containerised service was operating correctly.

Resource monitoring was performed using htop and free -m while the nginx container was running. Linux Lite recorded the lowest memory consumption at approximately 1313 MB, while Xubuntu and Lubuntu recorded approximately 1402 MB and 1437 MB respectively. CPU utilisation remained low across all three distributions, generally fluctuating between 1% and 4% during normal operation.

From a usability perspective, all distributions remained responsive while hosting the nginx service. Lubuntu appeared particularly lightweight and responsive during navigation, while Xubuntu provided a more feature-rich desktop environment at the cost of slightly higher resource usage. Linux Lite demonstrated efficient memory usage and maintained stable performance throughout testing.

The successful deployment of Docker and nginx on all three operating systems confirms their suitability for hosting lightweight containerised services in a small-scale private cloud environment.

Week 4 – Performance Evaluation and Analysis

During Week 4, the focus shifted from deployment to performance evaluation and comparative analysis of the three selected Linux distributions: Lubuntu, Xubuntu, and Linux Lite. The objective was to determine how effectively each operating system could support lightweight private cloud workloads while maintaining usability and system responsiveness.

The performance data collected during Weeks 2 and 3 was analysed and compared. Key metrics included idle memory usage, boot times, resource consumption during Docker-

based nginx deployment, and overall system responsiveness. These measurements provided an indication of how each distribution performed both under normal operating conditions and while hosting a lightweight containerised service.

Analysis of the baseline measurements showed that Linux Lite recorded the lowest idle memory usage at approximately 892 MB. Ubuntu recorded 958 MB, while Xubuntu recorded 998 MB. Startup performance testing indicated that Ubuntu achieved the fastest boot time at approximately 13 seconds, followed by Xubuntu at approximately 16 seconds. Linux Lite recorded the slowest startup time at approximately 42 seconds.

Additional analysis was conducted while running the Docker-based nginx service. Linux Lite again demonstrated the lowest memory consumption, using approximately 1313 MB of RAM during operation. Xubuntu consumed approximately 1402 MB, while Ubuntu consumed approximately 1437 MB. CPU utilisation remained low across all distributions during testing, indicating that the nginx workload placed minimal processing demands on the systems.

Usability observations were also recorded throughout testing. Ubuntu provided a highly responsive desktop environment with minimal resource overhead due to its LXQt interface. Xubuntu offered a more polished and feature-rich user experience through the XFCE desktop environment, although this resulted in slightly higher resource consumption. Linux Lite provided a familiar and accessible interface while maintaining strong memory efficiency during workload testing.

No significant stability issues were encountered during performance testing. All three distributions successfully hosted the Docker container and nginx service without crashes, service failures, or unexpected interruptions. This demonstrated that each distribution was capable of supporting lightweight private cloud services within the virtualised testing environment.

The findings from Week 4 indicate that Linux Lite achieved the best memory efficiency, Ubuntu delivered the fastest startup performance, and Xubuntu provided the most feature-rich desktop experience. These results will be used in the final stages of the project to evaluate the suitability of each distribution for small-scale private cloud deployment and to support the conclusions presented in the final IEEE paper.